DATE

Part 3: Gradient Descent Manual Code.

Given the linear equation:

$$y = m_1 x_1 + m_2 x_2 + b$$

Written in Matrix-format:

$$y = Xm + b$$

Where: $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$b = 1$ or broadCasted for later

$$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

DATE

Computing the prediction using Matrix

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$b = [1], \quad m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = X \cdot m + b$$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

~~$\hat{y} = \left((1 \times -1) + (3 \times 2) \right)$~~

$$\hat{y} = \begin{bmatrix} (1 \times -1) + (3 \times 2) \\ (4 \times -1) + (10 \times 2) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

DATE

Visualizing in the table

x_1	x_2	y	$\hat{y}$	$e=\hat{y}-y$	e^2
1	3	5	6	1	1
4	10	6	17	11	121

$$MSE = \frac{1}{n}(e_1^2 + e_2^2)$$

Where $n=2$, $e_1^2 + e_2^2 = 121+1$

$= 122$

$$MSE = \frac{1}{2}(122)$$

$$MSE = 61$$

DATE

Update #1

Previous Values:

$$m_0 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad b_0 = [1]$$

finding: $m_1 = ?$ and $b_1 = ?$

$$m_1 = m_0 - \alpha \frac{\partial J}{\partial m_0}$$

Where J is Cost function

$$J(m,b) = e^2, \text{ where, } e = \hat{y} - y$$

$$\text{therefore, } J(m,b) = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 / n$$

and $\hat{y} = Xm + b$

DATE

$$J(m,b) = \sum_{i=1}^{n} (Xm+b-y_i)^2 / n$$

$\frac{\partial J}{\partial m}$; - Using Chain rule

let $Xm+b-y$ be e

$$e = Xm+b-y$$

$$\frac{\partial J}{\partial m} = \frac{1}{n} \sum_{i=1}^{n} (Xm+b-y)^2$$

~~$\frac{\partial J}{\partial m} = \frac{\partial J}{\partial m}$~~

$$\frac{\partial J}{\partial m} = \frac{1}{n} \sum_{i=1}^{n} (e)^2$$

DATE

$$\frac{\partial J}{\partial m} = \left[\frac{\partial J}{\partial e} \cdot \frac{\partial e}{\partial m}\right] \frac{1}{n} \sum_{i=1}$$

$$\frac{\partial J}{\partial e} = 2e$$

$$\frac{\partial e}{\partial m} = X$$

$$\frac{\partial J}{\partial m} = \frac{1}{n} 2e \cdot X^T$$

Where X^T - X transpose

e - $\hat{y} - y$

$n = 2$

$$\frac{\partial J}{\partial m} = \frac{1}{2} \cdot 2(\hat{y} - y) \cdot X^T = X^T \cdot (\hat{y} - y)$$

DATE

therefore $\boxed{\frac{\partial J}{\partial m} = X^T(\hat{y} - y)}$

$$\frac{\partial J}{\partial b} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y} - y)^2$$

$$\hat{y} = Xm + b$$

$$e = Xm + b - y$$

$$\frac{\partial J}{\partial b} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\partial J}{\partial e} \cdot \frac{\partial e}{\partial b}\right]$$

$$= \frac{1}{n}\sum_{i=1}^{n}\left(2(\hat{y} - y) \cdot 1\right)$$

$$= \frac{1}{2}\sum_{i=1}^{n}\left(2(\hat{y} - y)\right)$$

$$= \sum_{i=1}^{n}(\hat{y} - y)$$

DATE

Therefore, $\frac{\partial J}{\partial b} = \sum_{i=1}^{n} (\hat{y}_i - y_i)$

Let learning rate be $\alpha = 0.1$

$$m_1 = m_0 - \alpha \cdot \frac{\partial J}{\partial m}$$

$$b_1 = b_0 - \alpha \cdot \frac{\partial J}{\partial m}$$

Where $m_0 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \quad X^1 = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

DATE

$$m_1 = m_0 - \alpha(X^T(\hat{y}-y))$$

$$X^T \cdot (\hat{y}-y) = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$m_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.1 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$m_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 4.5 \\ 11.3 \end{bmatrix}$$

$$m_1 = \begin{bmatrix} -5.5 \\ -9.3 \end{bmatrix}$$

I

DATE

$$b_1 = b_0 - \alpha \frac{\partial J}{\partial b}$$

$$b_1 = b_0 - \alpha \left(\sum_{i=1}^{n} (\hat{y}_i - y_i) \right)$$

$$\sum_{i=1}^{n} (\hat{y} - y) = 1 + 11 = 12$$

$$b_1 = b_0 - \alpha (12)$$

Where $\alpha = 0.1$, $b_0 = 1$

$$b_1 = 1 - 0.1 \times 12$$

$$b_1 = 1 - 1.2$$

$$b_1 = -0.2$$

DATE

The new updated parameters ~~for~~ after iteration 1 are:

$$W_1 = \begin{bmatrix} -5.5 \\ -9.3 \end{bmatrix} \quad \& \quad b_1 = [-0.2]$$

Iteration 2

Given,

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, \quad \alpha = 0.1, \quad m_1 = \begin{bmatrix} -5.5 \\ -9.5 \end{bmatrix}, \quad b_1 = -0.2$$

$$\Rightarrow \hat{y} = X m_1 + b_1 \quad \Rightarrow \hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -5.5 \\ -9.3 \end{bmatrix} + \begin{bmatrix} -0.2 \\ -0.2 \end{bmatrix}$$

~~First row,~~

First row,

$$\hat{y}_1 = (1x - 5.5) + (3x - 9.3) - 0.2$$
$$= -5.5 - 27.9 - 0.2$$
$$\boxed{\hat{y}_1 = -33.6}$$

Second row,

$$\hat{y}_2 = (4x - 5.5) + (10x - 9.3) - 0.2$$
$$= -22 - 93 - 0.2$$
$$\boxed{\hat{y}_2 = -115.2}$$

$$\Rightarrow \boxed{\hat{y} = \begin{bmatrix} -33.6 \\ -115.2 \end{bmatrix}}$$

$$\Rightarrow \text{Error } (e) = \hat{y} - y$$

$$e = \begin{bmatrix} -33.6 \\ -115.2 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\boxed{e = \begin{bmatrix} -38.6 \\ -121.2 \end{bmatrix}}$$

Scanned with CamScanner

$$\Rightarrow MSE = \frac{1}{h}(e_1^2 + e_2^2)$$

$$MSE = \frac{1}{2}\left((-38.6)^2 + (-121.2)^2\right)$$

$$= \frac{1}{2}(1489.96 + 14689.44)$$

$$= \frac{16179.4}{2}$$

$$\boxed{MSE = 8089.7}$$

$$\Rightarrow \frac{\partial J}{\partial m} = X^T(\hat{y} - y)$$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\Rightarrow \frac{\partial J}{\partial m} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -38.6 \\ -121.2 \end{bmatrix}$$

First value,

$$\Rightarrow (1 \times -38.6) + (4 \times -121.2)$$

$$= -523.4$$

Second value,

$$\Rightarrow (3 \times -38.6) + (10 \times -121.2)$$

$$= -1327.8$$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} -523.4 \\ -1327.8 \end{bmatrix}$$

Scanned with CamScanner

$$\Rightarrow m_2 = m_1 - \alpha \frac{\partial J}{\partial m}$$

$$m_2 = \begin{bmatrix} -5.5 \\ -9.3 \end{bmatrix} - 0.1 \begin{bmatrix} -523.4 \\ -1327.8 \end{bmatrix}$$

$$= \begin{bmatrix} -5.5 \\ -9.3 \end{bmatrix} - \begin{bmatrix} -52.34 \\ -132.78 \end{bmatrix}$$

$$\boxed{m_2 = \begin{bmatrix} 46.84 \\ 123.48 \end{bmatrix}}$$

$$\Rightarrow \frac{\partial J}{\partial b} = \Sigma (\hat{y} - y)$$

$$\frac{\partial J}{\partial b} = -38.6 + (-121.2)$$

$$\frac{\partial J}{\partial b} = -159.8$$

$$\Rightarrow b_2 = b_1 - \alpha \frac{\partial J}{\partial b}$$

$$b_2 = -0.2 - 0.1(-159.8)$$

$$\boxed{b_2 = 15.78}$$

$$\Rightarrow m_2 = \begin{bmatrix} 46.84 \\ 123.48 \end{bmatrix}, \quad b_2 = 15.78$$

Scanned with CamScanner

Date / /

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \quad Y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} 46,84 \\ 123,48 \end{bmatrix}, \quad b_2 = 15,78$$

Step 1: Compute the prediction

- $\hat{Y} = X m_2 + b_2$
- $\hat{Y}_1 = (1)(46,84) + (3)(123,48) + 15,78$
  $= 46,84 + 370,44 + 15,78$
  $= 433,06$
- $\hat{Y}_2 = 4(46,84) + (10)(123,48) + 15,78$
  $= 187,36 + 1234,8 + 15,78$
  $= 1437,94$

$$\hat{Y} = \begin{bmatrix} 433,06 \\ 1437,94 \end{bmatrix}$$

Scanned with CamScanner

Step 2: Compute the errors

Date

$$e = \hat{y} - y$$

$$= \begin{bmatrix} 433,06 \\ 1431,94 \end{bmatrix} - \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 428,06 \\ 1431,94 \end{bmatrix}$$

Step3: Gradient with respect

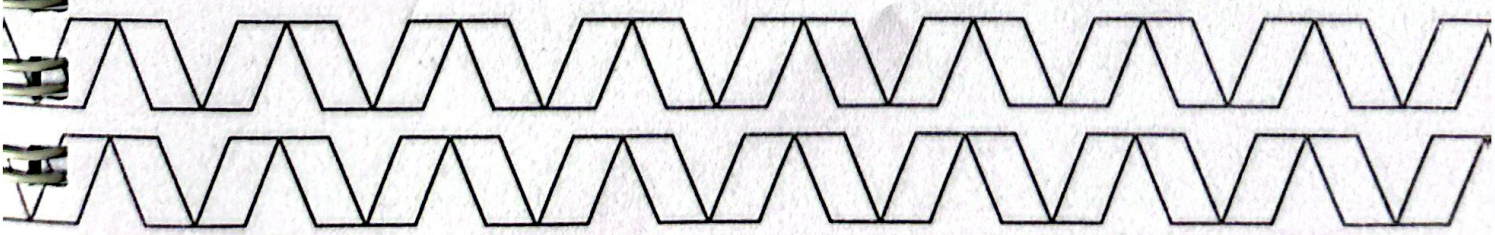
$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

first component

$$1(428,06) + 4(1431,94) = 6155,82$$

Second Component

$$3(428,06) + 10(1431,94) = 15502,58$$



Scanned with CamScanner

Date / /

$$\frac{\partial J}{\partial m} = \begin{bmatrix} 6155,82 \\ 1560,58 \end{bmatrix}$$

Step 4: Update m

$$m_3 = m_2 - \alpha \frac{J}{m}$$

$$= \begin{bmatrix} 46,84 \\ 123,48 \end{bmatrix} - 0,1 \begin{bmatrix} 6155,82 \\ 1560,58 \end{bmatrix}$$

$$= \begin{bmatrix} 46,84 \\ 123,48 \end{bmatrix} - \begin{bmatrix} 615,582 \\ 156,058 \end{bmatrix}$$

$$m_3 = \begin{bmatrix} -568,742 \\ -1436,778 \end{bmatrix}$$

Step 5: Gradient with b

$$\frac{\partial J}{\partial b} = \sum (\hat{y} - y)$$

$$= 428,06 + 1431,94$$

$$= 1860$$

Date / /

Step 6 update b

$b_3 = 15,78 - 0,1\ (1866)$

$= 15,78 - 186$

$= -170,22$

final answer

$$m_3 = \begin{bmatrix} -568,742 \\ -1436,778 \end{bmatrix}$$

$b_3 = -170,22$

Scanned with CamScanner

Iteration 4:

$$\hat{y} = Xm_3 + b_3$$

$$m_3 = \begin{bmatrix} -568.742 \\ -1436.77 \end{bmatrix}$$

$$b_3 = -170.22$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

1st Comp

$$(1)(-568.74) + (3)(-1436.78) + (-170.22)$$

$$= -568.74 - 4310.34 - 170.22 = -5049.30$$

2nd Comp

$$(4)(-568.74) + (10)(-1436.78) + (-170.22)$$

$$= -2274.96 - 1436.80 - 170.22 = -16812.98$$

Error(e) $= \hat{y} - y$

$$e = \begin{bmatrix} -5049.30 \\ -16812.98 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -5054.30 \\ -16818.98 \end{bmatrix}$$

$$MSE = \frac{1}{n}(e_1^2 + e_2^2)$$

$$MSE = \frac{1}{2}\left((-5054.30)^2 + (-16818.98)^2\right)$$

$$= \frac{1}{2}(25,545,948.49 + 282,878,088.24)$$

$$= 154,212,018.37$$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

1st Con

$$(1)(-5054.30) + (4)(-16818.98)$$

$$= -72330.22$$

2nd Con

$$(3)(-5054.30) + (10)(-16818.98)$$

$$= -183352.70$$

$$m_4 = m_3 - 0.1\begin{bmatrix} -72330.22 \\ -183352.70 \end{bmatrix} = \begin{bmatrix} -568.74 + 7233.02 \\ -1436.18 + 18335.27 \end{bmatrix}$$

$$m_4 = \begin{bmatrix} 6664.28 \\ 16898.49 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = \sum (y' - y)$$

$$-5054.30 + (-16818.98) = -21873.28$$

$$m$$

$$b_4 = -170.22 - 0.1(-21873.28)$$

$$b_4 = 2017.11$$

$$m_4 = \begin{bmatrix} 664.28 \\ 16898.49 \end{bmatrix}$$

$$b_4 = 2017.11$$